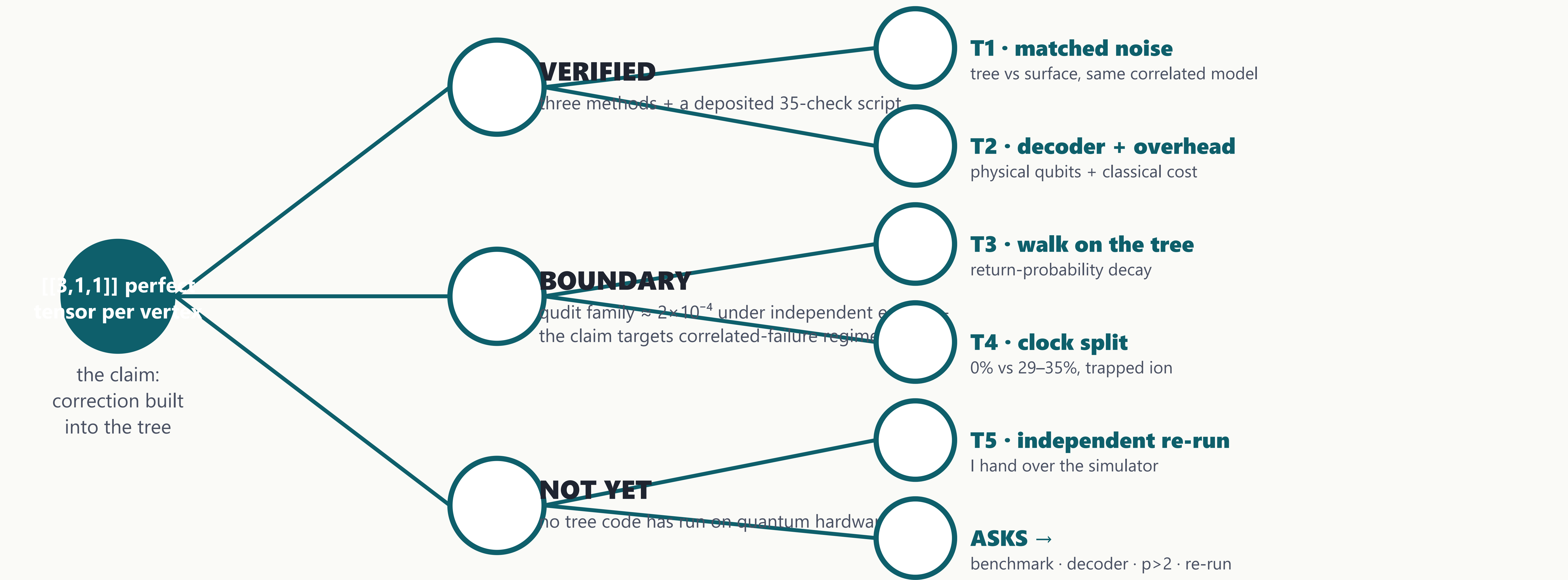


The Bruhat–Tits Tree Code — Status and Test Plan

Rowan Brad Quni-Gudzinās · QNFO · CWI Summer School on Quantum Algorithms & QEC · Wed 26 Aug 2026, 16:30–18:00



The tree is the code: the claim at the root, where it stands on the three branches, the five tests that would change my mind on the leaves.

50.0% bit-flip 4.6× surface (~10.9%)	75.0% depolarizing 75× surface (~1.0%)	17.30% independent X+Z no direct surface figure
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The tests

T1 • Matched-noise comparison
Tree vs surface under the *same* correlated noise model. Classical, days–weeks.
Would change my mind: the advantage falls to parity.

T2 • Decoder + full overhead
Physical qubits plus classical decoding cost. Classical.
Would change my mind: the advantage disappears once decoding cost is counted.

T3 • Random walk on a p-regular tree
Return-probability decay. Ion / analog simulator, ~8 weeks.
Would change my mind: no change in the decay exponent where predicted.

T4 • Clock-rest split
Diagonal 0% vs non-diagonal 29–35% violation rate. Trapped ion, ~8 weeks.
Would change my mind: the measured split falls outside that range.

T5 • Independent re-run
An outside group repeats T1 with my simulator and the protocol.
Would change my mind: nothing to predict — this one is about reproducibility.

For this room

- A correlated-noise benchmark — model and data — I can run T1 against.
- A decoder reality check: does a known decoder family already cover tree-structured codes, and how does it scale?
- Pointers on perfect tensors for $p > 2$ (the binary $[[3,1,1]]$ case is settled).
- An independent re-run of T1.

Verified: thresholds computed three independent ways — analytical recursion, exact stabilizer simulation, Monte Carlo — and re-checked by a deposited 35-check script.

Boundary, published: under independent errors the qudit family sits at $\approx 2 \times 10^{-4}$, about 55× below surface codes; the advantage claim targets correlated-failure regimes — which is exactly what T1 probes.